



UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education
Advanced Subsidiary Level and Advanced Level

MATHEMATICS

9709/04

Paper 4 Mechanics 1 (M1)

May/June 2007

1 hour 15 minutes

Additional Materials: Answer Booklet/Paper
Graph Paper
List of Formulae (MF9)

READ THESE INSTRUCTIONS FIRST

If you have been given an Answer Booklet, follow the instructions on the front cover of the Booklet.

Write your Centre number, candidate number and name on all the work you hand in.

Write in dark blue or black pen.

You may use a soft pencil for any diagrams or graphs.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

Where a numerical value for the acceleration due to gravity is needed, use 10 m s^{-2} .

The use of an electronic calculator is expected, where appropriate.

You are reminded of the need for clear presentation in your answers.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

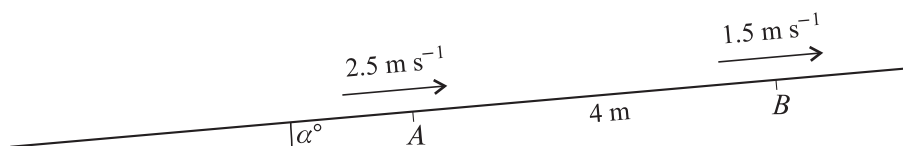
The total number of marks for this paper is 50.

Questions carrying smaller numbers of marks are printed earlier in the paper, and questions carrying larger numbers of marks later in the paper.

This document consists of 4 printed pages.



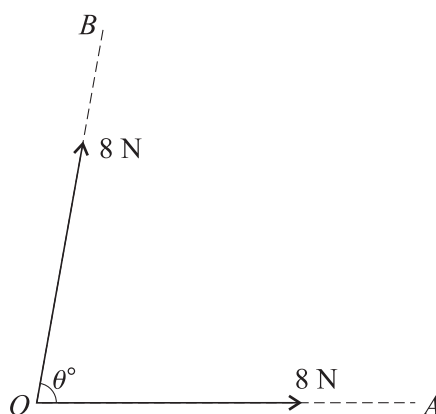
1



A particle slides up a line of greatest slope of a smooth plane inclined at an angle α° to the horizontal. The particle passes through the points A and B with speeds 2.5 m s^{-1} and 1.5 m s^{-1} respectively. The distance AB is 4 m (see diagram). Find

- (i) the deceleration of the particle, [2]
- (ii) the value of α . [2]

2



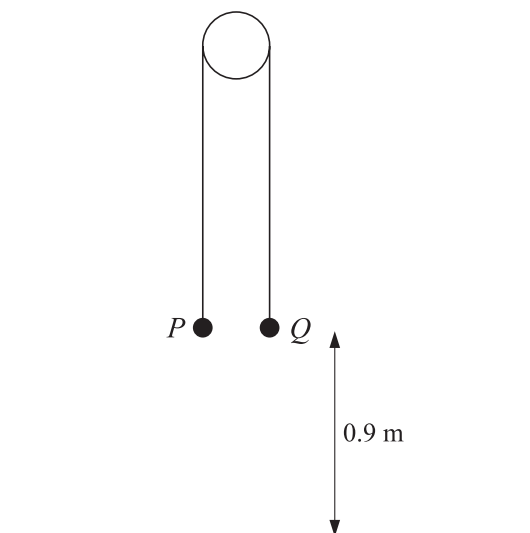
Two forces, each of magnitude 8 N , act at a point in the directions OA and OB . The angle between the forces is θ° (see diagram). The resultant of the two forces has component 9 N in the direction OA . Find

- (i) the value of θ , [2]
- (ii) the magnitude of the resultant of the two forces. [3]

3 A car travels along a horizontal straight road with increasing speed until it reaches its maximum speed of 30 m s^{-1} . The resistance to motion is constant and equal to $R \text{ N}$, and the power provided by the car's engine is 18 kW .

- (i) Find the value of R . [3]
- (ii) Given that the car has mass 1200 kg , find its acceleration at the instant when its speed is 20 m s^{-1} . [3]

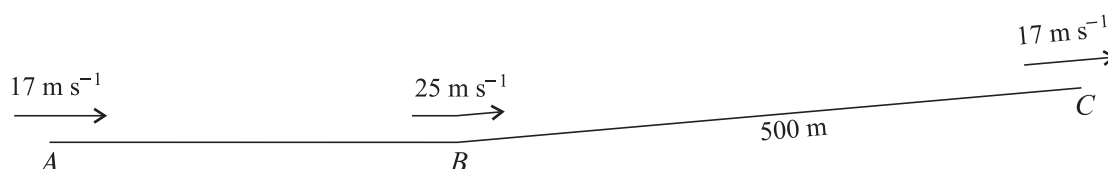
4



Particles P and Q , of masses 0.6 kg and 0.2 kg respectively, are attached to the ends of a light inextensible string which passes over a smooth fixed peg. The particles are held at rest with the string taut. Both particles are at a height of 0.9 m above the ground (see diagram). The system is released and each of the particles moves vertically. Find

- (i) the acceleration of P and the tension in the string before P reaches the ground, [5]
- (ii) the time taken for P to reach the ground. [2]

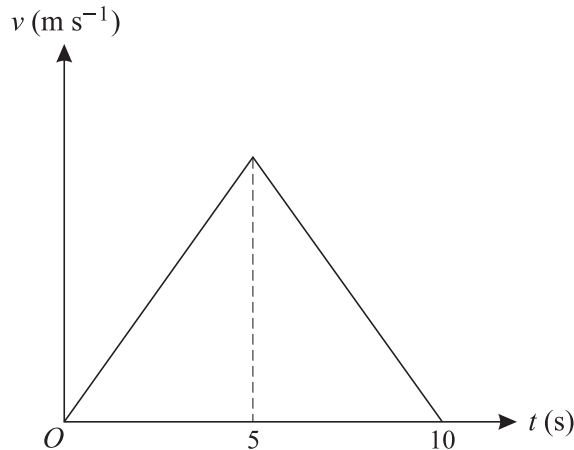
5



A lorry of mass $12\,500\text{ kg}$ travels along a road that has a straight horizontal section AB and a straight inclined section BC . The length of BC is 500 m . The speeds of the lorry at A , B and C are 17 m s^{-1} , 25 m s^{-1} and 17 m s^{-1} respectively (see diagram).

- (i) The work done against the resistance to motion of the lorry, as it travels from A to B , is 5000 kJ . Find the work done by the driving force as the lorry travels from A to B . [4]
- (ii) As the lorry travels from B to C , the resistance to motion is 4800 N and the work done by the driving force is 3300 kJ . Find the height of C above the level of AB . [4]

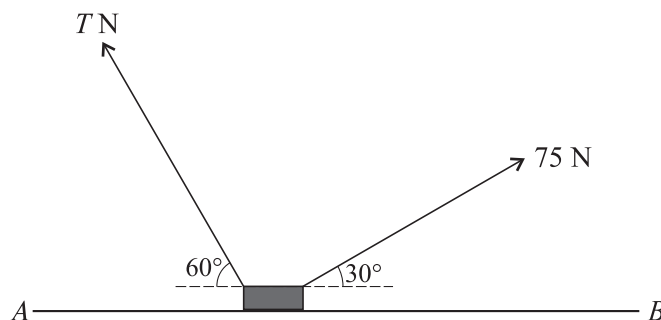
6



A particle P starts from rest at the point A and travels in a straight line, coming to rest again after 10 s. The velocity-time graph for P consists of two straight line segments (see diagram). A particle Q starts from rest at A at the same instant as P and travels along the same straight line as P . The velocity of Q is given by $v = 3t - 0.3t^2$ for $0 \leq t \leq 10$. The displacements from A of P and Q are the same when $t = 10$.

- (i) Show that the greatest velocity of P during its motion is 10 m s^{-1} . [6]
- (ii) Find the value of t , in the interval $0 < t < 5$, for which the acceleration of Q is the same as the acceleration of P . [3]

7



Two light strings are attached to a block of mass 20 kg. The block is in equilibrium on a horizontal surface AB with the strings taut. The strings make angles of 60° and 30° with the horizontal, on either side of the block, and the tensions in the strings are $T \text{ N}$ and 75 N respectively (see diagram).

- (i) Given that the surface is smooth, find the value of T and the magnitude of the contact force acting on the block. [5]
- (ii) It is given instead that the surface is rough and that the block is on the point of slipping. The frictional force on the block has magnitude 25 N and acts towards A . Find the coefficient of friction between the block and the surface. [6]

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